

IT2120 - Probability and Statistics

Lab Sheet 06

IT24103586 – Herath H.M.T.P

Exercise

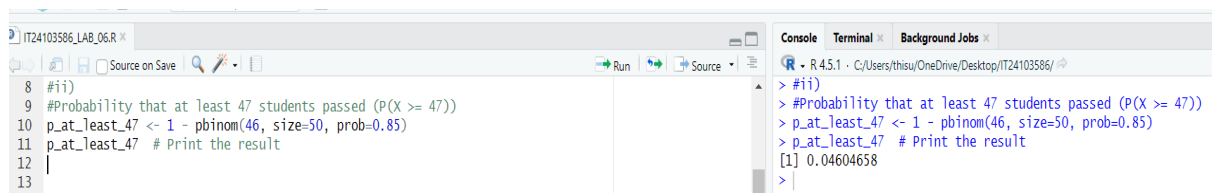
1. An IT company claims that their newly developed learning platform improves student performance in online tests. According to previous data, 85% of students who used the platform passed their online tests. A batch of 50 students is selected at random who have completed the course using this platform. Let X denote the number of students who passed the test out of 50 students.
 - i. What is the distribution of X ?

#Question 1

#i)

#Binomial Distribution with parameters $n=50$ and $p=0.85$

- ii. What is the probability that at least 47 students passed the test?



```
IT24103586_LAB_06.R x
Source on Save
Run Source
8 #ii)
9 #Probability that at least 47 students passed (P(X >= 47))
10 p_at_least_47 <- 1 - pbinom(46, size=50, prob=0.85)
11 p_at_least_47 # Print the result
12 |
13 |
Console Terminal Background Jobs x
R • R 4.5.1 • C:/Users/thisu/OneDrive/Desktop/IT24103586/
> #ii)
> #Probability that at least 47 students passed (P(X >= 47))
> p_at_least_47 <- 1 - pbinom(46, size=50, prob=0.85)
> p_at_least_47 # Print the result
[1] 0.04604658
> |
```

2. A call center receives an average of 12 customer calls per hour.

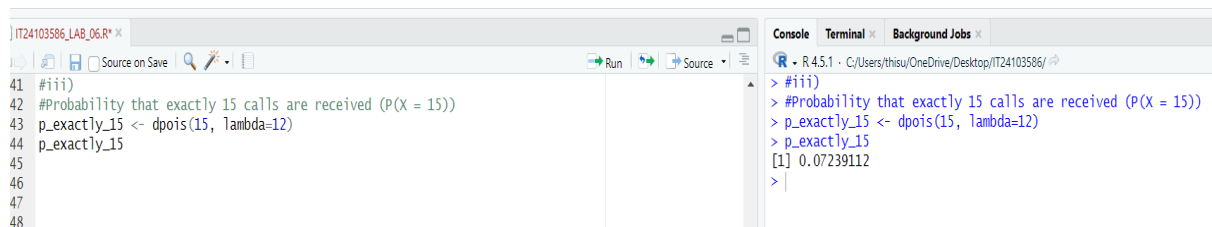
i. What is the random variable (X) for the problem?

```
#Question 2
#i)
#X is the number of customer calls received in an hour
"""
```

ii. What is the distribution of X?

```
#ii)
# Poisson distribution with parameter  $\lambda=12$  (average rate of calls per hour)
```

iii. What is the probability that exactly 15 calls are received in an hour?



The screenshot shows the RStudio environment. The script editor on the left contains the following R code:

```
41 #iii)
42 #Probability that exactly 15 calls are received (P(X = 15))
43 p_exactly_15 <- dpois(15, lambda=12)
44 p_exactly_15
45
46
47
48
```

The console on the right shows the execution of the code:

```
> #iii)
> #Probability that exactly 15 calls are received (P(X = 15))
> p_exactly_15 <- dpois(15, lambda=12)
> p_exactly_15
[1] 0.07239112
>
```